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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC602/20EC502 – COMPUTER NETWORKS

17EC701 - COMPUTER COMMUNICATION NETWORKS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe controlled access MAC protocols.	8	L*2	2	1
	b) Draw the seven-layer OSI model for computer networks. Summarize the responsibilities of each layer.	6	L2	1	1
	c) CRC method is used for coding the data words. Check if there is any error in the following messages. i) 1001110 ii) 1000110	6	L3	2	1
2.	a) With the help of flowchart explain the procedure of CSMA/CA protocol.	8	L2	2	1
	b) Write an example illustrate the use of physical logical, port and specific addressing methods in TCP/IP.	6	L2	1	1
	c) The following character encoding is used in a data link protocol: A: 01000111; B: 11100011; FLAG: 01111110; ESC: 11100000 Show the bit sequence transmitted (in binary) for the four-character frame: A B ESC FLAG when (i) Flag bytes with byte stuffing. (ii) Starting and ending flag bytes, with bit stuffing.	6	L3	2	1
3.	a) What are the three field types of frames in the HDLC protocol? Explain each of them briefly.	8	L2	1	1
	b) Explain stop and wait ARQ protocol with detailed design and necessary algorithms.	6	L2	2	1
	c) A Pure ALOHA network transmits 200-bit frames on a shared channel of 200kbps. What is the throughput if the system produces (a) 1000 frames per second (b) 500 frames per second (c) 250 frames per second.	6	L3	2	1
Unit – II					
4.	a) Describe the concept of VLAN in terms of membership and configuration. Also state the advantages of VLANs.	8	L2	3	1
	b) Explain link state routing protocol used to update the routing table.	6	L2	4	1
	c) What is backbone network? Explain the bus and star backbone networks in detail.	6	L2	3	1
5.	a) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows: i. The first group has 64 customers; each needs 256 addresses. ii. The second group has 128 customers; each needs 128 addresses. iii. The third group has 128 customers; each needs 64 addresses. Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.	8	L3	3	1

P.T.O.

- b) Explain IPv6 address in terms of their structure and address space.
 c) Describe three domains of domain name space. Also compare zone and domains.
6. a) Draw the TCP header format and brief the function of each field.
 b) Write the procedure for developing a routing table in distance vector routing method. Hence develop the routing table for the router A shown in Figure Q6(b).

6	L2	3	1
6	L2	4	1
8	L2	4	1

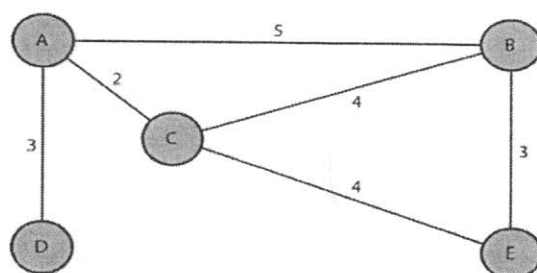


Figure Q6(b)

- c) List and explain any five connecting devices used in computer networks.

6	L3	4	1
6	L2	3	1

Unit – III

7. a) Explain Symmetric and Asymmetric cryptography methods.
 b) Explain the Packet-Filter Firewall and Proxy Firewall
8. a) Write short notes on i) Message integrity ii) IP Sec.
 b) Explain the classifications of cybercrimes.

	L2	5	1
10	L2	5	1
10	L2	5	1
10	L2	5	1
10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
